Materia: Programación de Sistemas Adaptativos

Actividad Fundamental No. : R.3.C.1 - Redes Neuronales (individual)

Nombre completo	**Matrícula**	**Carrera**	**Semestre**
Diego Enrique Vega Lara	2057220	ITS	6

Plan: 401	**Semestre de materia (Plan)**: 5
Salón: 1305	
Grupo: 002	**Día:** 30 / Abril / 2025
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Docente:	Sara Elena Garza Villarreal
Semestre Enero – Junio 2025	
San Nicolas de Los Garza, Nuevo León	

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Individual assignment : Neural Networks

I. Answer the exercises. Upload your answers to the specified location and attach, as well, a file with your procedure (or justification for your answer).

* Problem 1 (10 points). Label each formula depending on whether it is used for foward propagation (FP) or back propagation (BP).

(BP) a. $\partial w = \partial z * x$
(BP) b. $\partial b = \partial z$
(FP) c. $a = g(z)$
(FP) d. $z = w^T * x$
(BP) e. $\partial z = a - y$

Forward propagation (Activation calculation)
Backward propagation (weight update)

letter c. is activation
letter d. is core

* Problem 2. (10 points). Calculate db, considering that $\partial z = 0.7$

Gradient descent		
Gradient	Simplified notation	Formula
$\partial L / \partial z$	∂z	$\partial z = a - y$
$\partial L / \partial w$	∂w	$\partial w = \partial z * x$
$\partial L / \partial b$	∂b	$\partial b = \partial z$

$\partial b = \partial z$
if $\partial z = 0.7$
then $\underline{\underline{\partial b = 0.7}}$

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* Problem 3 (10 points). Calculate $a = R(z)$ (ReLU activation function), considering that $z = 0.58$

$a = R(z)$
$z = 0.58$

$\therefore R(z) \longrightarrow g(z) = R(z) \longrightarrow g(z) = \sigma(z)$

$$\longrightarrow \sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0.58}} = 0.641$$

I use the Sigmoidal actuation function instead, I didn't know how to solve this using $R(z) = \max(0, z)$.
And based on my procedure, I concluded that

$\sigma(z)$ iqual to $R(z)$

* Problem 4 (15 points). Indicate what $a_4^{[3]}$ denotes. Select the right option.

a. Denotes neuron 4 of example 3
b. Denotes neuron 3 of layer 4
c. Denotes neuron 4 of layer 3
d. Denotes the output of the neural network.

[illegible]

- [illegible] Feature j of the example is denoted as x_j^i, which "4" is the feature of a_4^3

- Example i is denoted as $x^{(i)}$, which "3" is the example of a_4^3

- The training set is denoted as x, which "a" is the ~~example of~~ training set of "a"

With this, I conclude that "a." is the answer.

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* Problem 5. (15 points). Update the value for b, considering that $\alpha = 0.01$, $\partial b = 0.5$, and currently $b = 1.3$

$\alpha = 0.01$

$\partial b = 0.5$

currently $b = 1.3$

Updating weights formulas

$w = w - \alpha * \partial w$

$b = b - \alpha * \partial b$

α · learning rate
Learning velocity

we are using this one

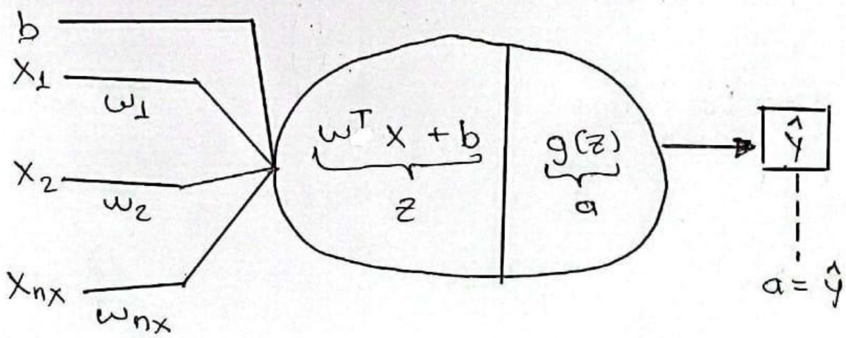
$$b = 1.3 - 0.01 * 0.5$$

$$\underline{\underline{b = 1.295}}$$

* Problem 6 (20 points). Calculate the error $L(a,y)$, considering that $y=1$, that the sigmoidal activation function is used, and that

$L(a,y)$; $y = 1$; $w = \begin{vmatrix} 0.5 \\ 0 \\ 1.5 \end{vmatrix}$ $x = \begin{vmatrix} 0.2 \\ 0.5 \\ 0.9 \end{vmatrix}$

Case: 1 Example / 1 Neuron



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$$z = \begin{vmatrix} 0.5 \\ 0 \\ 1.5 \end{vmatrix} * \begin{vmatrix} 0.2 \\ 0.5 \\ 0.9 \end{vmatrix} + b$$

I didn't know what to do for "b", so I tried to believe "b" will be 1.5

$$z = [\, 0.1 \quad 0 \quad 1.35 \,] + [1.5]$$

$$z = 2.95$$

$$\hat{y} = a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\hat{y} = \frac{1}{1 + e^{-2.95}}$$

$$\hat{y} = 0.95026$$

Sustituir en

$$L(\hat{y}, y) = -[y \ln \hat{y} + (1-y) \ln (1-\hat{y})]$$
$$L(\hat{y}, y) = \underline{0.20028}$$

* Problem 7. (20 points). Calculate dw, considering that $a = 1.2$, $y = 1$ and that

$$x = \begin{vmatrix} 0 \\ -1 \\ 1 \end{vmatrix}$$

We calculate "dz" first

$$\partial z = a - y$$
$$\partial z = 1.2 - 1$$
$$\partial z = 0.2$$

$$\partial w = \partial z * x$$

$$\partial w = \begin{vmatrix} 0.2 \\ 0.2 \\ 0.2 \end{vmatrix} * \begin{vmatrix} 0 \\ -1 \\ 1 \end{vmatrix} = \underline{\begin{vmatrix} 0 \\ -0.2 \\ 0.2 \end{vmatrix}} =$$